

Software & System Requirements

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Team Member / Author	Naga Lakshmi
Project Name	Smart Lender: AI-Based Loan Eligibility Prediction System
Document Type	System Requirements Specification

2.1 Software Requirements

- Operating System:** Windows 10/11, macOS Big Sur (or newer), or standard Linux distributions (Ubuntu/CentOS).
- Python Runtime:** Python versions 3.8 to 3.11 installed with Pip package manager configured.
- Web Browsers:** Modern browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, or Apple Safari with JavaScript enabled.
- Integrated Dev Env (IDE):** VS Code, PyCharm, or Jupyter Notebooks (for preliminary model exploration and parameter tuning).

2.2 Hardware Requirements

Hardware Component	Minimum Requirement	Recommended Requirement
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 (Dual-core or equivalent)	Intel Core i5 / AMD Ryzen 5 (Quad-core or better)
System Memory (RAM)	4 GB RAM	8 GB or 16 GB RAM (Highly recommended for executing heavy pipelines like XGBoost/RF)

Hardware Component	Minimum Requirement	Recommended Requirement
Storage Space	2 GB Free Disk Space (for Python interpreter, dependencies, and datasets)	10 GB Free SSD Space (for local environments, serialized pipelines, historical logs, and datasets)

3. System Workflow Dependencies

1. **Applicant Input / CSV Batch Upload** (via Web Interface/HTML5 Form)
2. **Data Processing Layer** (Pandas cleaning & Scikit-Learn pipelines)
3. **Inference Processing** (Model loading via Joblib, predictions via XGBoost/Random Forest)
4. **Output Delivery** (JSON payload back to UI, interactive dashboards, and downloadable PDF reports)